Clustering Hypergraph U-Net With High-Order Topology-Aware Pooling and Unpooling for Hyperspectral Image Classification

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***Abstract*—Recently, hypergraph convolutional networks (HGCNs) have shown considerable potential in hyperspectral image (HSI) classification for their superiority in capturing the complex many-to-many structure among multiple ground object samples. However, the absence of high-order pooling operation for many-to-many structure restricts networks to shallow architectures, limiting the extraction of deep discriminative features. To break this limitation, we propose a clustering hypergraph U-Net (CHGUnet) incorporating high-order topology-aware pooling and unpooling operations. Specifically, a sparse fuzzy clustering (SFC) strategy is first designed to explore multivariate relationships among multiple superpixels of ground objects, thereby building the high-order topological association hypergraph for hierarchical hypergraph convolution in the U-Net architecture. Meanwhile, the hypergraph pooling (HGPool) and hypergraph unpooling (HGUnPool), designed with high-order topology-aware assessments on both views of node and hyperedge, are adapted for adjusting high-order topological association to learn more suitable structured features that are beneficial to classification. Finally, a lightweight depth separable convolutional branch is integrated into the network through a cross-level interactive fusion (CLIF) mechanism, enhancing the superpixel-level features derived from hypergraph convolution with complementary pixel-level details. Such a CHGUnet can learn comprehensive discriminative features from a high-order topological perspective, and the experiments on three HSI datasets have proven its competitive performance in contrast to other advanced methods.**

***Index Terms*—Graph convolution, graph U-Net, high-order topology, hypergraph learning, hyperspectral images (HSIs) classification.**

I. Introduction

RECENTLY, the hyperspectral image (HSI) has garnered significant attention in the field of remote sensing [1]. Departing from traditional panchromatic or color images, HSI captures hundreds of nanoscale spectral resolution wavebands of the Earth's surface, which can be regarded as adding spectral dimensions to the 2-D images [2], [3]. The enormous abundance of spectral information brings not only the opportunity for refined land cover classification but also results in complex distribution and redundant information [4]. Therefore, discovering the potential structural property of data and mining effective features from HSI is a critical issue discussed by researchers in HSI classification [5], [6].

With the rapid development of deep learning theory, the deep neural network has been proposed to automatically learn deep abstract features and perform HSI classification tasks under the end-to-end architecture [7], [8]. Zhang et al. [9] divided each HSI patch into diverse regions according to six different directions and proposed a diverse region CNN network (DRCNN). For collaboratively capturing information in spatial and spectral domains of HSI, a fused 3DCNN (F3DCNN) model [10], designed by fusing 3DCNN and 2DCNN, explores the association between spectral and spatial features. While CNN-based methods have demonstrated significant improvements in the HSI classification, their convolutional operators remain fundamentally constrained by fixed local receptive fields [11]. This architectural limitation inherently restricts their ability to capture nonlocal information between long-range spatial regions, which are crucial for accurate land-cover classification in heterogeneous environments [12], [13]. To address this problem, the graph convolutional network (GCN) developed by theories of graph learning and deep learning has also been tried for HSI classification [14]. Mou et al. [15] proposed a nonlocal GCN (NLGCN) network that focuses on the strongly correlated samples with long spatial distances to capture the global relevant information. In order to accelerate the network training, Hong et al. [16] sampled a series of subgraphs from the complete graph and achieved the minibatch GCN (miniGCN), in which the sample nodes from each subgraph are used for batch training. With the help of superpixel segmentation, Chen et al. [17] integrated residual connections into graph convolution and proposed a dual-residual graph convolutional network (DRGCN) to learn deeper structural spectral–spatial features of HSI.

Although existing GCNs have successfully extended convolutional learning to non-Euclidean data through pairwise graph [18], this point-to-point structural representation limits their ability to capture the complex high-order spatial–spectral interactions inherent in HSI [19], [20]. In HSI classification tasks, land-cover objects typically exhibit both the groupwise spectral correlations and the multiregion spatial dependencies

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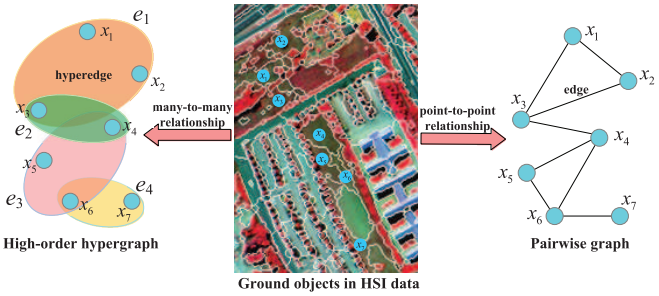


Fig. 1. Comparison of the pairwise graph and high-order hypergraph.

[21], and these characteristics cannot be effectively represented by simple pairwise adjacency matrices [22]. With the evolution of graphs, hypergraph have been developed to represent the high-order relationship by setting hyperedges between multiple nodes [23], generally greater than 2, as shown in Fig. 1. Based on the high-order structure in hypergraph, a few hypergraph convolutional networks (HGCNs) were attempted to HSI classification recently. Ma et al. [24] generated hypergraphs under views of different modalities and proposed a feature fusion hypergraph neural network (F2HNN) to capture the high-order structural multimodal features in HSI. In [25], a dynamic hypergraph convolutional network (DHCN) is designed by updating the hypergraph to model suitable high-order interactions among pixels and extract the global information of HSI. By leveraging hyperedges to model high-order topological structure of multiple targets, these HGCN methods conducting convolution on hypergraphs are more effective to capture groupwise interactive information for enabling more comprehensive feature representation, which surpasses pairwise limitations of traditional GCNs.

While HGCNs have demonstrated promising results for HSI classification, two fundamental challenges persist when they are applied to deeper architectures and complex land-cover scenarios. The first limitation stems from the over-smoothing phenomenon, where repeated graph propagation leads to homogenized node representations, progressively eroding feature discriminability in deeper network layers [26], [27]. Second, the inherent constraint of operating on fixed-scale topological structures limits the model's adaptability to multiscale land-cover patterns, preventing effective extraction of diverse fine-grained features across different spatial distributions [28]. Inspired by the popular U-Net architecture, some scholars attempted to introduce the encoder–decoder framework with learnable pooling and unpooling operations to build graph U-Net, enabling hierarchical multiscale learning while mitigating over-smoothing in deep GCN models [29], [30]. Guo et al. [31] proposed a dual-graph U-Net (DGUnet) that implements pooling and unpooling operations on spectral and spatial graphs. Furthermore, Liu et al. [32] developed a multilevel superpixel structured graph U-Net (MSSGU), where the graph pooling and unpooling operations established in the encoder–decoder framework enable features transfer from one scale graph to another, thereby enabling multiscale graphs to collaborate in one network. In addition, Qu et al. [33] further advanced this direction by integrating spatial and spectral attention into the graph U-Nets architecture to build the spatial–spectral attention graph U-Net (SSAGU) for generating subtler multiscales fusion features.

Current graph U-Nets predominantly employ pooling and unpooling operations designed for pairwise graphs, which prove inadequate for hypergraphs due to their distinctive high-order topology. Conventional graph pooling mechanisms are not applicable to hypergraphs that contain critical topological information not only in nodes but also in hyperedges. Furthermore, the lack of dedicated HGUnPool operations severely compromises the reconstruction of pooled hypergraph structures, ultimately limiting their effectiveness for HSI node classification. The absence of the pooling and unpooling operations paradigm for high-order topological structures fundamentally impedes the adaptation of U-Net architectures to HGCNs, restricting high-order topological features aggregation and redistribution in encoder–decoder frameworks. In addition, prevailing HGCN implementations typically construct hypergraphs by establishing hyperedges between individual samples and their immediate neighbors, which merely captures localized intraneighborhood correlations rather than the genuine multivariate relationships that characterize meaningful high-order structures in the HSI data.

To tackle the above problems, a clustering hypergraph U-Net (CHGUnet) with high-order topology-aware hypergraph pooling and unpooling (HGPool and HGUnPool) is proposed. In this method, a sparse fuzzy clustering (SFC) strategy is first designed to explore superpixels of ground objects with common properties, and the multivariate relationships between multiple clusters are exploited to describe the high-order associations of ground objects in the hypergraph. Then, based on the hypergraph convolution, a hierarchical hypergraph U-Net integrated with a depthwise separable convolutional auxiliary branch is built to extract the complementary structured features on the superpixel level and pixel level. More importantly, in the encoding and decoding stages of CHGUnet, a pair of high-order topology-aware HGPoo and deviation feature guided HGUnPool operations are designed to realize the compression and redistribution of hypergraph scale under the guidance of high-order topological correlation, which helps to enable hierarchical multiscale learning while mitigating over-smoothing effects in deep network layers. In summary, the main contributions of this article are listed as follows.

1) Unlike most hypergraphs just considering the neighborhood structure of each sample, the hypergraph built on the proposed SFC strategy represents the high-order topological structure through adaptively exploring the multivariate relationships among multiple superpixel sample clusters, which ensures convolution on the hypergraph for extracting the high-order structural features via the many-to-many topological associations in HSI.
2) For the special high-order structure in hypergraph, HGPool removes the worthless hyperedges and fully aggregated nodes by the high-order topology-aware assessments on both views of node and hyperedge, which obviously alleviates the over-smoothing issues encountered in deep hypergraph convolutional layers.

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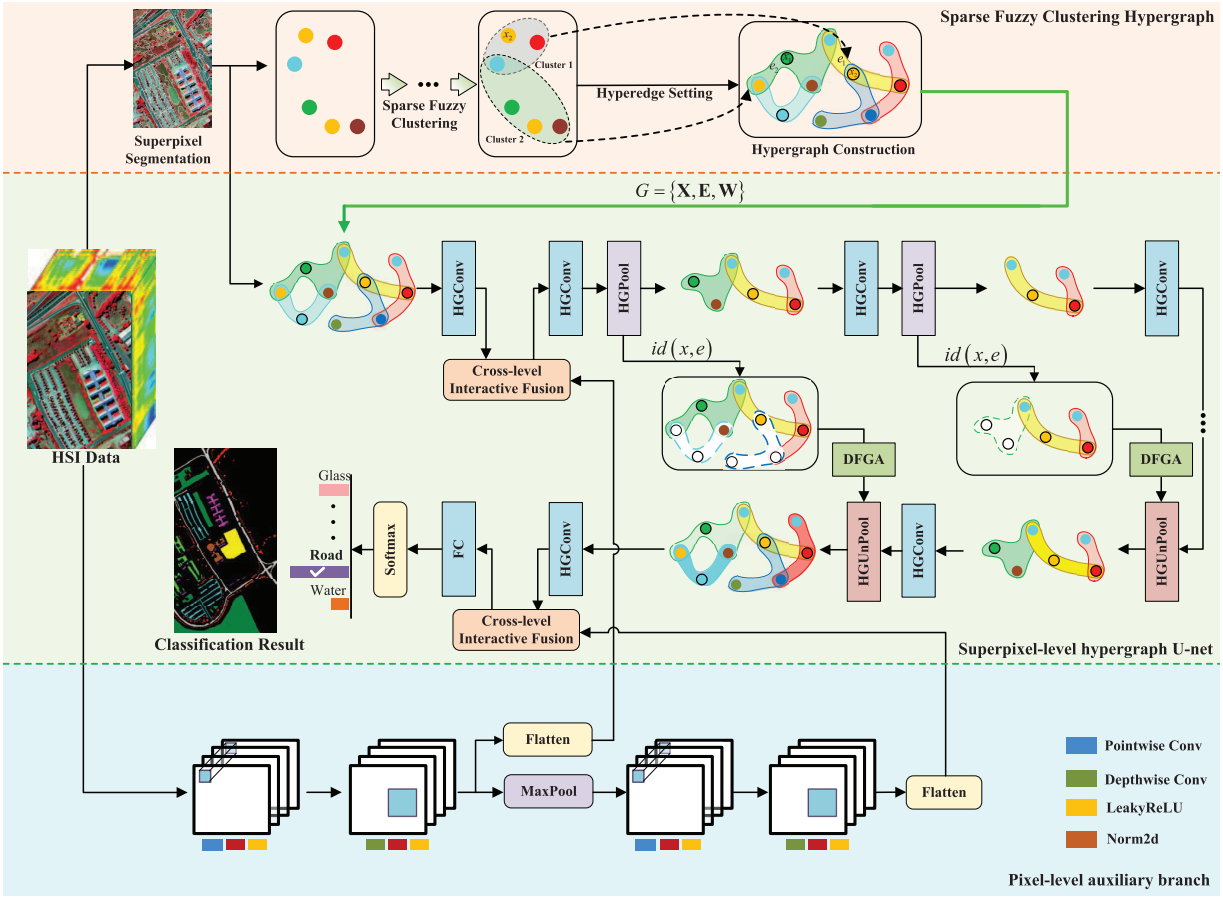


Fig. 2. Framework of the proposed CHGUnet method. The entire framework employs a dual-branch architecture where the hypergraph U-Net captures structural features at the superpixel level while the lightweight depthwise separable branch extracts fine-grained semantic features at the pixel level. The integrated CLIF module realizes the complementary representation of the two-level features.

3) The HGUnPool is conducted on the pooled hypergraph via the deviation feature guidance attention between the corresponding layers in the encoding and decoding stages, which not only restores the topology information lost in previous HGPool layers but also optimizes the hypergraph weights for subsequent structural convolution.
4) The lightweight depthwise separable convolution focusing on pixel-level feature learning is integrated into the hypergraph U-Net architecture via a cross-level interactive fusion (CLIF) mechanism, thereby enhancing superpixel-level features derived from hypergraph convolution with complementary pixel-level details for fine HSI classification.

The remaining sections of this article is organized as follows. Section II gives the details about the proposed method. The classification results, parameter analysis, and ablation experiments are presented in Section III. Section IV draws the conclusion and outlooks.

II. Proposed Method

The overall framework of the proposed CHGUnet is shown in Fig. 2. After constructing the hypergraph via the SFC on superpixels, CHGUnet is established with a main stage of hierarchical hypergraph U-Net including a pair of novel HGPool and HGUnPool operations on the hypergraph. Meanwhile, an auxiliary branch consisting of two depthwise separable convolutional blocks is integrated into the hypergraph U-Net architecture by the CLIF mechanism, which bridges the gap between superpixel-level and pixel-level features for promoting comprehensive structured representation for HSI.

A. Hypergraph Construction With SFC

In this work, an original HSI is denoted as $\mathbf{X}_o \in \Re^{H\times W\times B}$, where H and W denote its spatial sizes and B represents the number of bands, respectively. The label of the pixel is represented as $l \in \{1, 2, \ldots, C\}$, and C is the number of classes. Given that the hypergraph built on pixel-level will lead to too large graph sizes and bring a huge computational burden, the simple linear iterative clustering (SLIC) [34] algorithm is utilized to segment HSI data into $s = (H \times W)/m$ superpixels and build a superpixel sample set $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_s]$, where sample $\mathbf{x}_i$ is the mean of the pixels contained in the ith superpixel. m is the average number of pixels in each superpixel. Treating each superpixel sample as a node is an effective strategy to reduce the hypergraph size.

Aiming to achieve convolution driven by high-order topological structure, the multivariate relationships among

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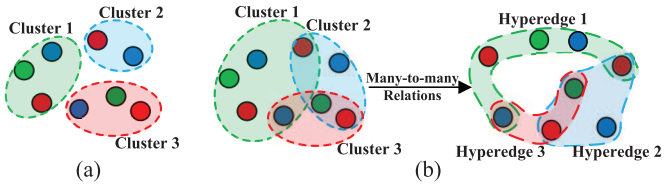


Fig. 3. High-order hypergraph construction. (a) K-means hard clustering. (b) SFC.

superpixel samples need to be considered in hypergraph construction. Inspired by the unsupervised clustering strategy that can reveal the relationships between multiple samples in different grouping clusters, we propose an SFC method relying on sparse representation and fuzzy clustering theories to uncover multivariate correlations among multiple overlapping clusters, thereby enabling the high-order hypergraphs construction, as shown in Fig. 3.

SFC first randomly selects some initial samples from $\mathbf{X}$ as the centroids of clusters as $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_s]$, and then uses fuzzy membership degree u_{ij} to indicate that the sample $\mathbf{x}_i$ belongs to a certain category $\mathbf{v}_j$. The number of clusters is $c = s/C$. To find the optimal cluster centers and membership degree that minimizes the weighted distance from all samples to cluster centers, the objective function of SFC is defined as follows:

$$\begin{cases} J = \underbrace{\sum_{i}^{s}\sum_{j}^{c} u_{ij}^2 \left\|\mathbf{x}_i - \mathbf{v}_j\right\|_2^2}_{\text{smooth}} + \lambda \underbrace{\sum_{i}^{s}\sum_{j}^{c} \left\|u_{ij}\right\|_1}_{\text{nonsmooth}} \\ \text{s.t} \quad \sum_{j}^{c} u_{ij} = 1 \end{cases} \tag{1}$$

where parameter λ controls the sparse degree of u_{ij}.

Benefit from the role of ℓ_1-norm sparse constraint on fuzzy membership degree u_{ij}, this objective function explores sparse correlations among all samples to effectively select sparse neighbors for fuzzy clustering, which enables fuzzy memberships between each sample $\mathbf{x}_i$ and a few clusters or even one single cluster is nonzero. For the sparse membership degree u_{ij}, its nonzero value indicates that sample $\mathbf{x}_i$ belongs to the jth cluster and reflects its importance in this cluster. The existence of constraint term $\sum_j^c u_{ij} = 1$ ensures that each sample $\mathbf{x}_i$ belongs to at least one cluster. For the smooth term in objective function, its Lagrange function is

$$L = \sum_{i}^{s}\sum_{j}^{c} u_{ij}^2 \left\|\mathbf{x}_i - \mathbf{v}_j\right\|^2 - \alpha\left(\left(\sum_{j}^{c} u_{ij}\right) - 1\right). \tag{2}$$

By setting the partial derivatives of membership degree u_{ij} and multipliers α as 0, u_{ij} can be obtained by

$$u_{ij} = 1 \Big/ \sum_{k=1}^{c}\left(\frac{\left\|\mathbf{x}_i - \mathbf{v}_j\right\|}{\left\|\mathbf{x}_i - \mathbf{v}_k\right\|}\right)^2. \tag{3}$$

Then, a soft threshold function is applied to the obtained u_{ij} to introduce the sparsity of nonsmooth regularization as follows:

$$u_{ij} = \operatorname{sign}\left(u_{ij}\right)\max\left(0, |u_{ij}| - \lambda\right). \tag{4}$$

With the obtained u_{ij}, the centroid of cluster is updated by

$$\mathbf{v}_j = \frac{\sum_{i=1}^{s} u_{ij}\mathbf{x}_i}{\sum_{i=1}^{s} u_{ij}}. \tag{5}$$

The above clustering iteration will terminate when samples contained in each cluster no longer change. Based on the membership degree u_{ij} obtained by SFC, the high-order hypergraph $G = \{\mathbf{X}, \mathbf{E}, \mathbf{W}\}$ can be constructed, in which $\mathbf{E} = \{e_1, e_2, \ldots, e_c\}$ stands for the set of hyperedges, and $\mathbf{W} = \operatorname{diag}\{w(e_j)\} \in \Re^{c\times c}$ is a diagonal matrix and the diagonal element represents the weight of each hyperedge. The hyperedge $e_j \in \mathbf{E}$ is set to connect all samples in the jth cluster, so the incidence matrix $\mathbf{H} = [h(\mathbf{x}_i, e_j)]_{i,j}$ that denotes the connection situation of samples and hyperedges is defined as follows:

$$h\left(\mathbf{x}_i, e_j\right) = \begin{cases} 1, & u_{ij} \neq 0 \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

While the weight $w(e_j) \in \mathbf{W}$ of hyperedge e_j is defined as follows:

$$\begin{aligned} w\left(e_j\right) &= \exp\left(-\frac{\sum_{\mathbf{x}_i\in e_j} u_{ij}\left\|\mathbf{x}_i - \bar{x}_j\right\|^2}{\sum_{e_j\in\mathbf{E}}\sum_{\mathbf{x}_i\in e_j} u_{ij}\left\|\mathbf{x}_i - \bar{x}_j\right\|^2 / |\mathbf{E}|}\right) \\ &= \exp\left(-\frac{\operatorname{tr}\left(\mathbf{M}^{we}\right)}{\operatorname{tr}\left(\mathbf{M}^{be}/|\mathbf{E}|\right)}\right) \end{aligned} \tag{7}$$

where $|\mathbf{E}|$ is the number of the hyperedges in hyperedge set. $\bar{\mathbf{x}}_j$ is the mean value of samples contained in hyperedge e_j. $\mathbf{M}^{we}$ and $\mathbf{M}^{be}$ are within-hyperedge and between-hyperedge sparse weighted scatter matrices that are calculated as follows:

$$\begin{cases} \mathbf{M}^{we} = \sum_{i=1}^{s} u_{ij}\left(\mathbf{x}_i - \bar{\mathbf{x}}_j\right)\left(\mathbf{x}_i - \bar{\mathbf{x}}_j\right)^{\mathrm{T}} \\ \mathbf{M}^{be} = \sum_{i=1}^{s}\sum_{j=1}^{c} u_{ij}\left(\mathbf{x}_i - \bar{\mathbf{x}}_j\right)\left(\mathbf{x}_i - \bar{\mathbf{x}}_j\right)^{\mathrm{T}}. \end{cases} \tag{8}$$

The traces of $\mathbf{M}^{we}$ and $\mathbf{M}^{be}$ indicate the comprehensive similarity of multiple samples within the jth hyperedge or between different hyperedges. With the constructed hypergraph G and the corresponding incidence matrix $\mathbf{H}$, the degree matrix of the node in the hypergraph is $\mathbf{D}_x = \operatorname{diag}\{d(\mathbf{x}_i)\} \in \Re^{s\times s}$, while the hyperedge degree matrix is represented by $\mathbf{D}_e = \operatorname{diag}\{d(e_j)\} \in \Re^{c\times c}$, which are diagonal matrices and their values are computed by

$$d\left(e_j\right) = \sum_{i=1}^{s} h\left(\mathbf{x}_i, e_j\right), \quad d\left(\mathbf{x}_i\right) = \sum_{i=1}^{c} h\left(\mathbf{x}_i, e_j\right) w\left(e_j\right). \tag{9}$$

The hypergraph convolution can be considered as a high-order extension of traditional graph convolution. Mathematically, driven by the spectral graph theory [24], the hypergraph convolution operation (HGConv) can be expressed as follows:

$$\operatorname{HGConv}\left(G, \mathbf{X}, \Theta\right) = \sigma\left(\mathbf{D}_x^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^{\mathrm{T}}\mathbf{D}_x^{-1/2}\mathbf{X}\Theta\right). \tag{10}$$

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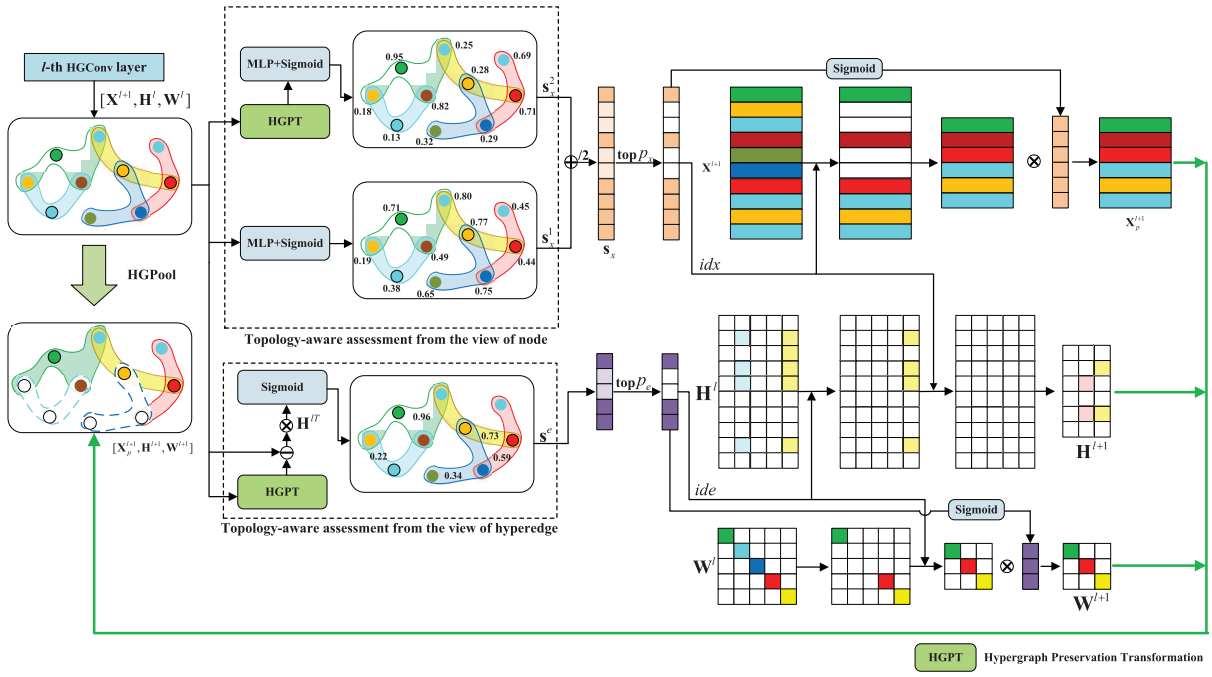


Fig. 4. Illustration of the HGPool layer. The topology-aware assessments are conducted from both views of the node and the hyperedge in the hypergraph.

B. Hypergraph U-Net via High-Order Topology-Aware HGPool

Building upon the initial clustering hypergraph via SFC, we establish a superpixel-level hypergraph U-Net to extract deep structured features under the high-order topological guidance, whose architecture is shown in Fig. 2. The hypergraph U-Net comprises $2P$ layers in total, with P layers each in the encoding and decoding stages. After each HGConv layer in the encoding stage, HGPool is implemented to compress the hypergraph for removing redundant hyperedges and nodes to avoid the over-smoothing phenomenon, and adjusting the topological structure to extract deeper structured features. The topology-aware assessments from both views of node and hyperedge are conducted to form HGPool layer, as shown in Fig. 4.

1) Topology-Aware Assessment From the View of Node: The hypergraph generally contains some saturation nodes that have been fully aggregated with their adjacent nodes after multilayer HGConv. It is necessary to select the important nodes requiring further convolutional operation, which can avoid the overfitting phenomenon on saturation nodes. Regarding the topological information among nodes, we consider the node status before and after the aggregation under the current topological structure to evaluate which nodes should be retained in the pooling. For the output $\mathbf{X}^{l+1}$ and hypergraph G^l of the lth HGConv layer, a multiple-layer perceptron (MLP) is first utilized to assess the importance of the current node state as follows:

$$\mathbf{s}_x^1 = \text{sigmoid}\left(\text{MLP}\left(\mathbf{X}^{l+1}\right)\right) \tag{11}$$

where MLP with learning parameters adaptively computes the current node feature assessment score $\mathbf{s}_x^1$.

To further assess the future node states, we remove learnable parameters Θ from the HGConv layer to form a hypergraph preservation transformation (HGPT) operation that performs node aggregation under the current topological structure

$$\text{HGPT}\left(\mathbf{X}^{l+1}\right) = \left(\mathbf{D}_x^l\right)^{-1}\mathbf{H}^l\mathbf{W}^l\left(\mathbf{D}_e^l\right)\mathbf{H}^{lT}\mathbf{X}^{l+1} \tag{12}$$

in which each node is aggregated with its adjacent nodes within the same hyperedge. When a node still has high importance after aggregation with the current hypergraph structure, this node should be considered for preservation in HGPool. The assessment score $\mathbf{s}_x^2$ of future nodes' state after HGPT is computed by

$$\mathbf{s}_x^2 = \text{sigmoid}\left(\text{MLP}\left(\text{HGPT}\left(\mathbf{X}^{l+1}\right)\right)\right). \tag{13}$$

If a node has significant importance before and after transformation, it indicates that this node needs to be preserved for subsequent hypergraph convolution. Thus, the assessment score of nodes is $\mathbf{s}_x = (\mathbf{s}_x^1 + \mathbf{s}_x^2)/2$.

Then, we rank the nodes according to their score $\mathbf{s}_x$ and select a subset of nodes to form $\mathbf{X}_p^{l+1}$ as follows:

$$\begin{cases} idx = \text{toprank}\left(\mathbf{s}_x, p_x\right) \\ \tilde{\mathbf{s}}_x = \text{sigmoid}\left(\mathbf{s}_x\left(idx\right)\right) \\ \mathbf{X}_p^{l+1} = \mathbf{X}^{l+1}\left(idx, :\right) \otimes \text{diag}\left(\tilde{\mathbf{s}}_x\right) \end{cases} \tag{14}$$

where p_x is the node pooling ratio, $\otimes$ is the matrix multiplication. diag() converts score vector $\mathbf{s}_x$ to diagonal matrix, while toprank() returns the indices of top p_x percent nodes. $\mathbf{s}_x(idx)$ and $\mathbf{X}^{l+1}(idx, :)$ realize the element or row extraction to form a new score vector and node matrix.

2) Topology-Aware Assessment From the View of Hyperedge: Regarding the topological features of hyperedges, it assumes that if nodes from one hyperedge own significant differences before and after the HGPT transformation, this hyperedge is considered as valuable adjacency information for

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subsequent convolution. Otherwise, it should be removed in pooling due to the existence of this edge having no significant effect on the updates of nodes. Based on this assumption, the difference of all nodes before and after HGPT within each hyperedge is used to evaluate the corresponding hyperedge as follows:

$$\begin{aligned}\mathbf{s}_e &= \text{sigmoid}\left(\mathbf{H}^{\mathrm{T}}\left(\mathbf{X}^{l+1} - \text{HGPT}\left(\mathbf{X}^{l+1}\right)\right)\mathbf{1}\right)\\ &= \text{sigmoid}\left(\mathbf{H}^{\mathrm{T}}\left(\mathbf{I} - \mathbf{D}_x^{-1}\mathbf{H}^l\mathbf{W}^l\mathbf{D}_e^{-1}\mathbf{H}^{lT}\right)\mathbf{X}^{l+1}\mathbf{1}\right)\end{aligned} \tag{15}$$

where $\mathbf{1} \in \Re^{B\times 1}$ is all-one vector and B is the dimension of node feature. $(\mathbf{X}^{l+1} - \text{HGPT}(\mathbf{X}^{l+1}))\mathbf{1}$ is a 1-D vector that represents the feature difference of all nodes before and after transformation. $\mathbf{s}_e$ is the assessment score vector of a hyperedge whose element is computed by the sum of differences between nodes within the corresponding hyperedge.

Besides the above node pooling, the hyperedge pooling is defined with the aid of both hyperedge score $\mathbf{s}_e$ and node score $\mathbf{s}_x$, and it is realized by dealing with incidence matrix $\mathbf{H}^l$ and weight matrix $\mathbf{W}^l$ as follows:

$$\begin{cases} ide = \text{toprank}\left(\mathbf{s}_e, p_e\right)\\ \mathbf{H}^{l+1} = \mathbf{H}^l\left(idx, ide\right)\\ \tilde{\mathbf{s}}_e = \text{sigmoid}\left(\mathbf{s}_e\left(idx\right)\right)\\ \mathbf{W}^{l+1} = \mathbf{W}^l\left(ide, ide\right) \otimes \text{diag}\left(\tilde{\mathbf{s}}_e\right)\end{cases} \tag{16}$$

where p_e is the hyperedge pooling ratio. $\mathbf{H}^l(idx, ide)$ and $\mathbf{W}^l(ide, ide)$ are functions that extract the element according to indexes to update $\mathbf{H}^{l+1}$ and $\mathbf{W}^{l+1}$ for pooled hypergraph.

Relying on the above node and hyperedge pooling, the lth HGPool layer outputs the pooled hypergraph G^{l+1} with new $\mathbf{X}_p^{l+1}$, $\mathbf{H}^{l+1}$, and $\mathbf{W}^{l+1}$ for next $(l+1)$th HGConv layer as follows:

$$\mathbf{X}^{l+2} = \sigma\left(\left(\mathbf{D}_x^{l+1}\right)^{-1}\mathbf{H}^{l+1}\mathbf{W}^{l+1}\left(\mathbf{D}_e^{l+1}\right)\mathbf{H}^{l+1T}\mathbf{X}_p^{l+1}\Theta^{l+1}\right) \tag{17}$$

where σ is the activation function and Θ^{l+1} is the learnable parameter. $\mathbf{X}^{l+2}$ is the output of $(l+1)$th HGConv layer.

C. Deviation Features Guided HGUnPool

Opposite to the HGPool layer, the HGUnPool layer based on deviation features guidance attention (DFGA) is proposed to restore the topology structure of the previous hypergraph in the decoding stage of the U-Net architecture. Fig. 5 shows the processing steps of the $(l+i)$th HGUnPool layer that matches the lth HGPool layer. Relying on the positions of the selected nodes and hyperedges in the corresponding lth HGPool layer, we relocate them to the original hypergraph structure in the matching $(l+i)$th HGUnPool layer. The unpooling operations for nodes, hyperedges, and weight are defined as follows:

$$\begin{cases}\mathbf{X}_{\text{up}}^{l+i+1} = \text{distribute}\left(\mathbf{X}^{l+1}\left(idx, :\right), \mathbf{X}^{l+i+1}\right)\\ \mathbf{H}^{l+i+1} = \text{distribute}\left(\mathbf{H}^{l+1}\left(idx, ide\right), \mathbf{H}^{l+i}\right)\\ \mathbf{W}_{\text{up}}^{l+i+1} = \text{distribute}\left(\mathbf{W}^{l+1}\left(ide, ide\right), \mathbf{W}^{l+i}\right)\end{cases} \tag{18}$$

where idx and ide are indexes of the selected nodes and hyperedges in the lth HGPool layer. distribute() arranges the row vectors in $\mathbf{X}^{l+i+1}$ or elements in $\mathbf{H}^{l+i}$ and $\mathbf{W}_{\text{up}}^{l+i+1}$ into the previous matrix according to indexes idx and ide.

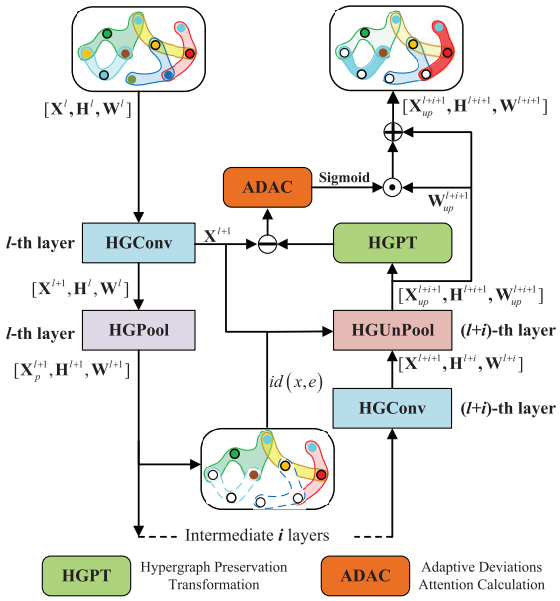


Fig. 5. DFGA for HGUnPool layer.

Given that some weights in $\mathbf{W}_{\text{up}}^{l+i+1}$ from previous lth layer may not be suitable for the current node status of $\mathbf{X}^{l+i+1}$, the DFGA that considers the deviation features of nodes from the pairwise HGPool-HGUnPool layers is designed to regulate the weights and emphasize the meaningful hyperedges. The deviation features are obtained by the elementwise subtraction as $\mathbf{X}^{l+1} - \text{HGPT}(\mathbf{X}^{l+i+1})$. HGPT operation is utilized to explore the structured information of the current unpooling hypergraph. Afterward, the deviation features are fed to calculate the attention weight by

$$\begin{aligned}&\text{Att}_{\text{DFGA}} =\\ &\qquad\text{sigmoid}\left(\mathbf{H}^{(l+i+1)T}\left(\mathbf{X}^{l+1} - \text{HGPT}\left(\mathbf{X}_{\text{up}}^{l+i+1}\right)\right)\Theta\right)\end{aligned} \tag{19}$$

where Θ is the hyperparameter in adaptive deviations attention calculation (ADAC) block. Each row of matrix $\mathbf{H}^{(l+i+1)T}(\mathbf{X}^{l+1} - \text{HGPT}(\mathbf{X}_{\text{up}}^{l+i+1}))$ is the sum of the deviation features of all nodes connected in the corresponding hyperedges. While the final optimized weight matrix $\mathbf{W}^{l+i+1}$ is calculated as follows:

$$\mathbf{W}^{l+i+1} = \left(\text{Att}_{\text{DFGA}} + \mathbf{I}\right) \odot \mathbf{W}_{\text{up}}^{l+i+1}. \tag{20}$$

With the above $\mathbf{X}_{\text{up}}^{l+i+1}$, $\mathbf{H}^{l+i+1}$, and $\mathbf{W}^{l+i+1}$ obtained in HGUnPool layer, the unpooling hypergraph G^{l+i+1} is completed for subsequent convolutional learning in U-Net architecture.

D. Cross-Level Interactive Fusion

Hypergraph U-Net as the backbone of the network learns the high-order structural features of HSI at the superpixel level. To learn the differentiated features of each pixel within one superpixel, a lightweight auxiliary branch based on the depthwise separable convolution (DSConv) [35] is integrated into the hypergraph U-Net via the designed CLIF mechanism,

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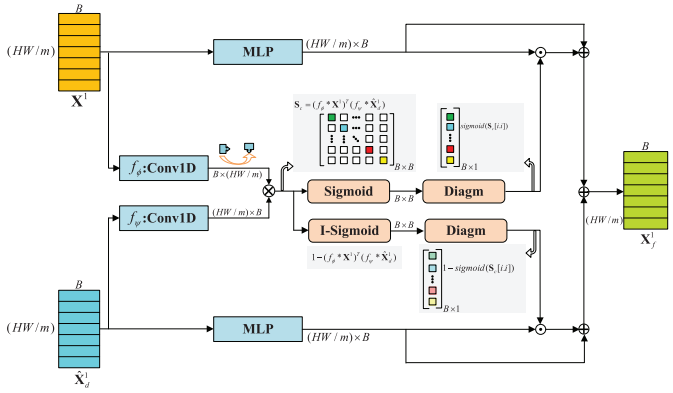


Fig. 6. CLIF mechanism.

for extracting the supplemental detailed features on pixel-level without increasing the computational load significantly. As shown in Fig. 2, there are two DSConv blocks in the auxiliary branch to extract the detailed semantics feature hidden in the local region of each pixel, and the outputs of the two DSConv blocks are integrated into the beginning of the encoding stage and the end of the decoding stage in the hypergraph U-Net, respectively. To achieve the fusion of superpixel feature $\mathbf{X} \in \Re^{(HW/m)\times B}$ and pixel-level feature $\mathbf{X}_d \in \Re^{H\times W\times B}$, a mapping association between pixel $\mathbf{x}_i^d$ and superpixel $\mathbf{x}_j$ is first designed by

$$\mathbf{O}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{x}_i^d \in S\left(\mathbf{x}_j\right) \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

where $S(\mathbf{x}_j)$ denote pixels set contained in superpixel $\mathbf{x}_j$.

With the mapping matrix $\mathbf{O} \in \Re^{HW\times(HW/m)}$, the pixel-level feature $\mathbf{X}_d^1$ of the first DSConv block and the superpixel feature $\mathbf{X}^1$ of the first HGConv layer are fused by

$$\mathbf{X}_f^1 = \text{CLIF}\left(\mathbf{X}^1, \hat{\mathbf{O}}^{\text{T}}\text{Flatten}\left(\mathbf{X}_d^1\right)\right) \tag{22}$$

where $\hat{\mathbf{O}}$ denotes the column normalization form and Flatten() transforms $\mathbf{X}_d$ from $H \times W \times B$ dimension to $HW \times B$ dimension. For convenience, we represent $\hat{\mathbf{X}}_d^1 = \hat{\mathbf{O}}^{\text{T}}\text{Flatten}(\mathbf{X}_d^1)$, where $\hat{\mathbf{X}}_d^1 \in \Re^{(HW/m)\times B}$. Considering the semantic differences between pixel-level features and superpixel-level features, CLIF() achieves the cross-level features fusion from the perspective of channels interactive attention. The structure of CLIF is shown in Fig. 6, in which the 1-D convolution is first performed to convert original features into another space where the number of channels in the levels feature is equal. And then the adaptive channels similarity of two-level features is calculated by inner product operation as follows:

$$\mathbf{S}_c = \left(f_\phi * \mathbf{X}^1\right)^{\text{T}}\left(f_\psi * \hat{\mathbf{X}}_d^1\right) \tag{23}$$

where f_ϕ and f_ψ are the convolutional kernels. $\mathbf{S}_c \in \Re^{d\times d}$ is the adaptive channel similarity matrix, in which each element $S_{c[i,j]}$ means the inner product of ith channel in $f_\phi * \mathbf{X}^1$ and jth channel in $f_\psi * \hat{\mathbf{X}}_d^1$.

To achieve better fusion for pairwise channels in superpixel-level and pixel-level features, two interactive attention weights Att_{sl} and Att_{pl} are calculated with the cross-level channel correlation as follows:

$$\begin{cases} \text{Att}_{\text{sl}} = \text{diagm}\left(\text{sigmoid}\left(\left(f_\phi * \mathbf{X}^1\right)^{\text{T}}\left(f_\psi * \hat{\mathbf{X}}_d^1\right)\right)\right) \\ \text{Att}_{\text{pl}} = \text{diagm}\left(\mathbf{I} - \text{sigmoid}\left(\left(f_\phi * \mathbf{X}^1\right)^{\text{T}}\left(f_\psi * \hat{\mathbf{X}}_d^1\right)\right)\right) \end{cases} \tag{24}$$

where operator diagm() extracts the diagonal elements of matrix to form a vector. After the sigmoid function, the adaptive channel similarity is rescaled in the range [0,1]. The sum of attention weights Att_{sl} and Att_{pl} is 1, which ensures the complementarity between features on superpixel and pixel levels. Then, the cross-level fusion feature is obtained by

$$\mathbf{X}_f^1 = \left(\mathbf{X}^1\Theta_1\right) \odot \left(\text{Att}_{\text{sl}} + \mathbf{I}\right) + \left(\hat{\mathbf{X}}_d^1\Theta_2\right) \odot \left(\text{Att}_{\text{pl}} + \mathbf{I}\right) \tag{25}$$

where Θ_1 and Θ_2 are the learning parameters in two MLP layers.

Similarly, in the end of hypergraph U-Net as shown in Fig. 2, superpixel feature $\mathbf{X}^l$ of the last hypergraph decoder is fused with pixel-level $\mathbf{X}_d^2$ of the second DSConv block as follows:

$$\mathbf{X}_f = \text{CLIF}\left(\hat{\mathbf{O}}\mathbf{X}^l, \text{Flatten}\left(\mathbf{X}_d^2\right)\right). \tag{26}$$

In the final output fused feature $\mathbf{X}_f \in \Re^{HW\times B}$, the pixel-level feature obtained by DSConv is a valuable supplement to the superpixel-level hypergraphs U-Net, which helps to achieve differentiated feature learning and fine classification of each individual pixel sample within the same superpixel.

After the linear transformation of fused feature in the fully connected (FC) layer, the softmax layer is utilized for predicting per-pixel class probabilities as follows:

$$\mathbf{Y}^{\text{pre}} = \text{soft max}\left(\text{FC}\left(\mathbf{X}_f\right)\right) \tag{27}$$

where $\mathbf{Y}^{\text{pre}}$ denotes the class probabilities of pixels in HSI. With the predictive result, the classic cross-entropy is adopted as classification loss, which is defined as follows:

$$\text{loss} = -\sum_{i=1}^{n}\sum_{j=1}^{c}\mathbf{Y}_{i,j}\log\left(\mathbf{Y}_{i,j}^{\text{pre}}\right) \tag{28}$$

where $\mathbf{Y}$ denotes the ground-truth label matrix.

III. Experimental Results and Analysis

A. Datasets

1) Indian Pines: This image was photographed by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, over the northwestern of Indiana. After the interference bands had been removed, 204 bands remained for experiments. The spatial size of this image is 145 × 145 and contains a total of 16 classes labeled pixels as shown in Table I.

2) PaviaU: This data were gathered at the University of Pavia, Italy, by the Reflective Optical System Imaging Spectrometer (ROSIS) sensor, which records 103 bands in the wavelength range of 430–860 nm. Nine categories are considered in this image with 610×340 pixels. The numbers of samples for each category and the randomly selected training samples are detailed in Table II.

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TABLE I
DETAILED INFORMATION OF INDIAN PINES DATASET

Color	ID	Class name	Total	Train	Test
	1	Alfalfa	46	23	23
	2	Corn-notill	1428	40	1388
	3	Corn-mintill	830	40	790
	4	Corn	237	40	197
	5	Grass-pasture	483	40	443
	6	Grass-trees	730	40	690
	7	Grass-pasture	28	14	14
	8	Hay-windrowed	478	40	438
	9	Oats	20	10	10
	10	Soybean-notill	972	40	932
	11	Soybean-mintill	2455	40	2415
	12	Soybean-clean	593	40	553
	13	Wheat	205	40	165
	14	Woods	1265	40	1225
	15	Buildings-Drives	386	40	346
	16	Stone-Steel	93	40	53
		Total	10249	567	9683

TABLE II
DETAILED INFORMATION OF THE PAVIAU DATASET

Color	ID	Class name	Total	Train	Test
	1	Asphalt	6631	40	6591
	2	Meadows	18649	40	18609
	3	Gravel	2099	40	2059
	4	Trees	3064	40	3024
	5	Painted metal	1345	40	1305
	6	Bare Soil	5029	40	4989
	7	Bitumen	1330	40	1290
	8	Self-Blocking	3682	40	3642
	9	Shadows	947	40	907
		Total	42776	360	42416

TABLE III
DETAILED INFORMATION OF THE HOUSTON DATASET

Color	ID	Class name	Total	Train	Test
	1	Healthy grass	1251	40	1211
	2	Stressed grass	1254	40	1214
	3	Synthetic grass	697	40	657
	4	Trees	1244	40	1204
	5	Soil	1242	40	1202
	6	Water	325	40	285
	7	Residential	1268	40	1228
	8	Commercial	1244	40	1204
	9	Road	1252	40	1212
	10	Highway	1227	40	1187
	11	Railway	1235	40	1195
	12	Parking Lot 1	1233	40	1193
	13	Parking Lot 2	469	40	429
	14	Tennis Court	428	40	388
	15	Running Track	660	40	620
		Total	15029	600	14429

3) Houston: The dataset was collected at the University of Houston, Houston, TX, USA, in a campus environment with the neighboring urban area. The image size is up to 349×1905, with 144 effective bands covering 380–1050 nm. Table III details information on samples across 15 land cover types.

B. Experimental Setup

1) Comparison Algorithms: To obtain convincing comparative results, a total of ten state-of-the-art algorithms have been used for comparison in experiments. Among them, DRCNN and F3DCNN are CNN-based networks. miniGCN, NLGCN, and DRGCN conduct various graph convolutional operations with different emphasis points, such as batch training modes or global nonlocal relations. MSSGU, DGUnet, and SSAGU are multiple versions of GCN U-Net with different pooling or unpooling strategies; DHCN and F2HNN belong to hypergraph-based networks focusing on the high-order structure of data.

2) Evaluation and Hyperparameter Setting: We conducted a series of classification experiments and realized the quantitative assessment with overall accuracy (OA), average accuracy (AA) and kappa coefficient. The values of the indicators listed in the experiments are the average of ten repetitions. The parameters of CHGUnet are updated by Adam optimizer with an initial learning rate of 0.0005, and the learning rate will drop to one-half after every 50 epochs. Sparse degree λ as a nonsensitive parameter in SFC for hypergraph construction is set to 0.1 according to experience. While other sensitive parameters, such as super-pixel partition scale, node, and hyperedge pooling rates, are discussed in detail in subsequent experiments. The hyperparameters of comparison approaches are in agreement with the corresponding references. All methods are implemented with Pytorch on a computer with an Nvidia A40 GPU.

C. Parameter Analysis

There are three key parameters in CHGUnet, including node pooling ratio p_x, hyperedge pooling ratio p_e, and average number m of pixels in each superpixel. p_x and p_e directly affect the simplification of topology information in the pooled hypergraph, while m controls the segmentation size of superpixel and then determines the scale of the hypergraph treating superpixels as nodes. We have explored their influence with different values and shown the corresponding results in Figs. 7 and 8.

1) Influence of Node and Hyperedge Pooling Ratios: As can be seen from Fig. 7, the average OAs tend to increase and then slightly decrease with the rise of p_x and p_e. The reason for this phenomenon is that larger ratios preserve more topology information in the pooling process for learning structured deep features. However, excessively large ratios will propagate ineffective topological associations to subsequent HGConv layers, causing excessive node aggregation and ultimately aggravating over-smoothing issues. In contrast to the hyperedge pooling ratio p_e, the node pooling ratio p_x generally exerts a stronger influence on classification performance. This difference becomes particularly pronounced when both parameters adopt smaller values. Based on these experimental results, the values of p_x and p_e were set to 0.85 and 0.8 for Indian Pines, 0.85 and 0.85 for PaviaU, 0.9 and 0.85 for Houston in subsequent experiments.

2) Influence of the Size of Superpixel: In HSIs, using individual pixels as nodes would create prohibitively large hypergraphs due to the massive number of pixels, leading to unacceptable computational and memory costs. Since neighboring pixels in homogeneous regions typically share similar spectral characteristics and land-cover categories, the superpixel approach effectively reduces redundancy by representing clusters of similar pixels with mean-pooled features as single nodes in the hypergraph. To minimize original pixel information loss caused by the introduction of the superpixel, a suitable superpixel size m was adopted to adapt to the distribution characteristics of ground objects, ensuring that pixels within the same superpixel have the same labels and characteristics. The curves of average OAs with respect to m

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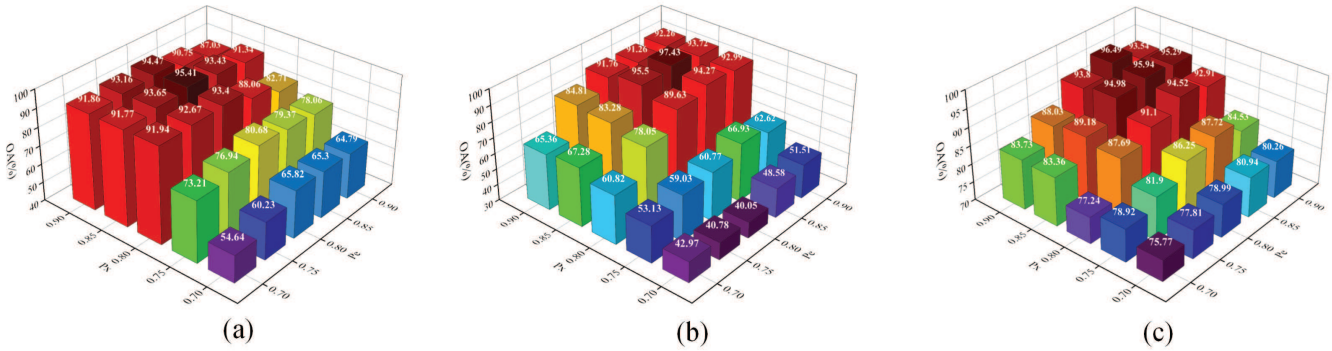


Fig. 7. OAs with different node and hyperedge pooling ratios on three datasets. (a) Indian Pines. (b) PaviaU. (c) Houston.

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class	DRCNN	F3DCNN	miniGCN	NLGCN	DRGCN	MSSGU	DGUnet	SSAGU	DHCN	F2HNN	CHGUnet
1	90.32±1.23	77.42±1.73	35.48±4.91	94.63±0.55	96.77±0.41	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**
2	70.32±3.72	75.62±1.75	50.00±3.36	83.11±1.30	80.19±2.11	90.13±1.26	68.80±2.85	95.37±0.52	85.78±1.71	93.28±0.71	**97.62±0.25**
3	70.38±3.33	82.69±1.33	46.41±4.43	67.34±3.23	73.85±1.90	95.68±0.51	80.00±2.05	89.32±1.06	82.95±1.37	**98.88±0.15**	86.78±1.51
4	90.37±0.86	86.63±1.26	89.30±0.90	66.31±3.73	93.58±0.51	92.63±0.52	93.58±0.61	94.81±0.64	95.79±0.30	94.20±0.66	**96.92±0.42**
5	94.23±0.73	95.15±0.41	92.38±0.71	96.93±0.24	96.07±0.52	91.79±0.93	92.15±0.89	98.47±0.18	95.61±0.46	82.78±1.27	**97.29±0.30**
6	96.91±0.32	97.94±0.28	95.88±0.41	99.56±0.04	99.85±0.01	95.63±0.44	98.82±0.13	99.72±0.04	99.56±0.06	95.29±0.42	**100±0.00**
7	100±0.00	100±0.00	76.92±1.49	92.86±0.63	92.31±0.76	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**
8	99.77±0.03	96.73±0.24	99.53±0.04	**100±0.00**	100±0.00	**100±0.00**	99.53±0.04	98.45±0.13	**100±0.00**	**100±0.00**	**100±0.00**
9	100±0.00	100±0.00	20.00±10.06	71.43±1.72	80.00±2.82	**100±0.00**	100±0.00	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**
10	72.23±2.90	75.05±3.52	68.98±1.85	73.89±3.47	79.72±2.40	77.31±2.18	72.34±3.91	85.88±0.31	80.91±2.24	85.56±1.88	**87.81±1.06**
11	58.09±4.63	62.12±3.59	74.64±3.47	83.85±2.19	82.99±1.73	73.22±2.52	66.40±2.42	81.98±1.08	69.40±3.50	93.15±0.77	**94.32±0.61**
12	74.22±2.77	86.74±0.88	54.33±5.47	60.58±4.94	88.58±1.11	89.88±1.29	79.14±1.38	87.32±1.53	92.08±0.82	82.42±1.57	**98.55±0.10**
13	**100±0.00**	99.35±0.05	98.06±0.19	98.71±0.08	98.71±0.08	98.92±0.07	**100±0.00**	**100±0.00**	**100±0.00**	98.86±0.08	**100±0.00**
14	83.21±1.39	95.06±0.46	92.84±0.76	98.03±0.16	93.33±0.77	99.04±0.07	88.07±0.88	98.79±0.12	94.57±0.62	**100±0.00**	98.45±0.21
15	80.06±1.94	88.10±1.29	59.23±3.16	66.32±2.89	89.58±0.63	99.73±0.02	88.99±1.09	**99.72±0.04**	96.73±0.24	97.19±0.28	98.84±0.15
16	**100±0.00**	97.67±0.19	**100±0.00**	91.49±0.98	**100±0.00**	98.63±0.12	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**
OA	80.06±2.58	85.46±1.58	74.16±3.61	86.39±0.94	88.74±1.15	91.33±1.55	84.84±2.38	93.46±0.83	90.46±1.36	93.88±0.70	**95.53±0.59**
AA	86.26±1.01	89.52±0.88	74.12±2.69	88.48±0.89	90.72±0.54	91.91±0.77	88.99±1.53	94.24±0.49	92.40±0.84	94.10±0.37	**96.06±0.23**
Kappa	78.25±1.44	82.36±0.96	71.49±2.34	82.89±0.68	85.43±0.87	88.74±1.32	79.61±1.92	90.36±0.77	86.62±0.65	92.48±0.63	**93.62±0.46**

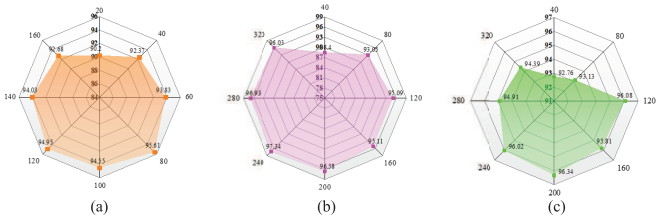


Fig. 8. OAs with respect to different superpixel sizes m. (a) Indian Pines. (b) PaviaU. (c) Houston.

are displayed in Fig. 8. In general, a larger m will lead to more samples contained in each superpixel. But an excessively large m will increase the probability of heterogeneous pixel samples being included in each superpixel region, leading to incorrect node feature representations. On the contrary, a parameter m that is too small will compress the superpixel size, leading to an increase in the number of nodes in the hypergraph and exacerbating the computational burden of HGConv. For Indian Pines dataset, the proposed method performs best performance when m = 80. While for PaviaU and Houston datasets with larger image sizes, the suitable m = 300 and 250.

D. Comparison of Classification Results

The classification accuracies of the proposed method and other compared methods on each class are listed in Tables IV–VI, along with OA, AA, Kappa, and classification accuracy in each class. Furthermore, the classification maps generated by different methods on all datasets are shown in Figs. 9–12.

As listed in Tables IV–VI, among CNN-based methods, F3DCNN obtains better classification accuracies than DRCNN due to its powerful spatial and spectral feature extraction capability of 3-D convolutional operation. NLGCN and DRGCN take the whole HSI image as input and perform nonlocal regions information propagation by graph convolutions. As for total of six GCN-based comparison methods, MSSGU and SSAGU produce better results thanks to their U-Net architecture designed on graph pooling and unpooling operations, which enables them to transfer features between multiscale graphs in different depth layers for learning deeper structural features. Although DGUnet also adopts the U-Net architecture, its accuracy is slightly lower than MSSGU and SSAGU due to the lack of mining structural features at the superpixel-level that uses common local regions as nodes for aggregation learning. DHCN and F2HNN, by using the high-order topology in hypergraph convolution, obtain generally better experimental results. Especially for F2HNN, besides the method proposed in this article, achieved the highest classification accuracy on Indian Pines and Houston datasets, indicating that high-order topological information is beneficial for extracting more discriminative structural features.

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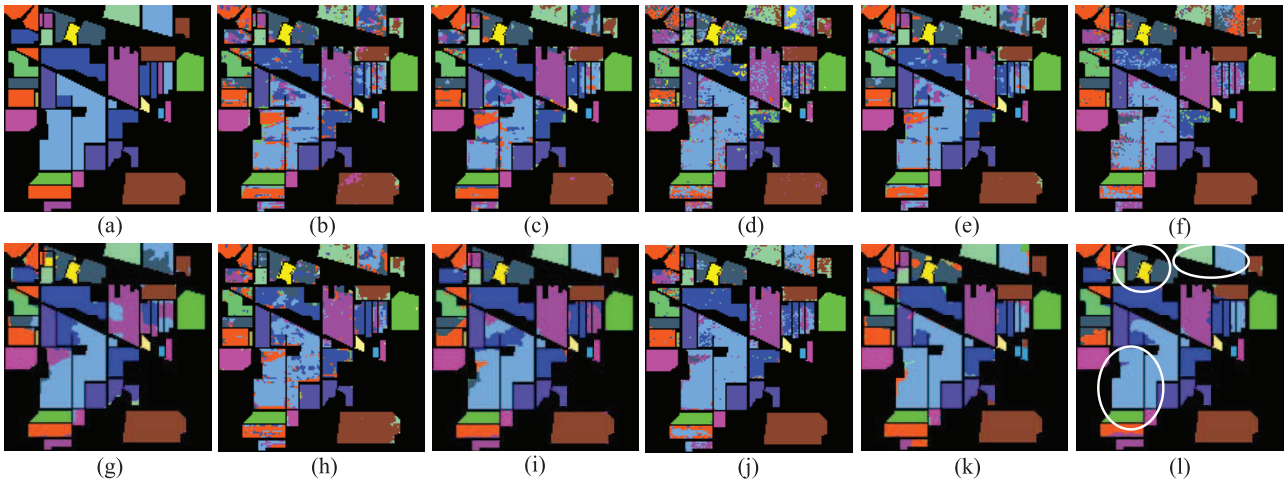


Fig. 9. Classification maps on Indian Pines dataset. (a) Ground-truth map. (b) DRCNN. (c) F3DCNN. (d) miniGCN. (e) NLGCN. (f) DRGCN. (g) MSSGU. (h) DGUnet. (i) SSAGU. (j) DHCN. (k) F2HNN. (l) CHGUnet.

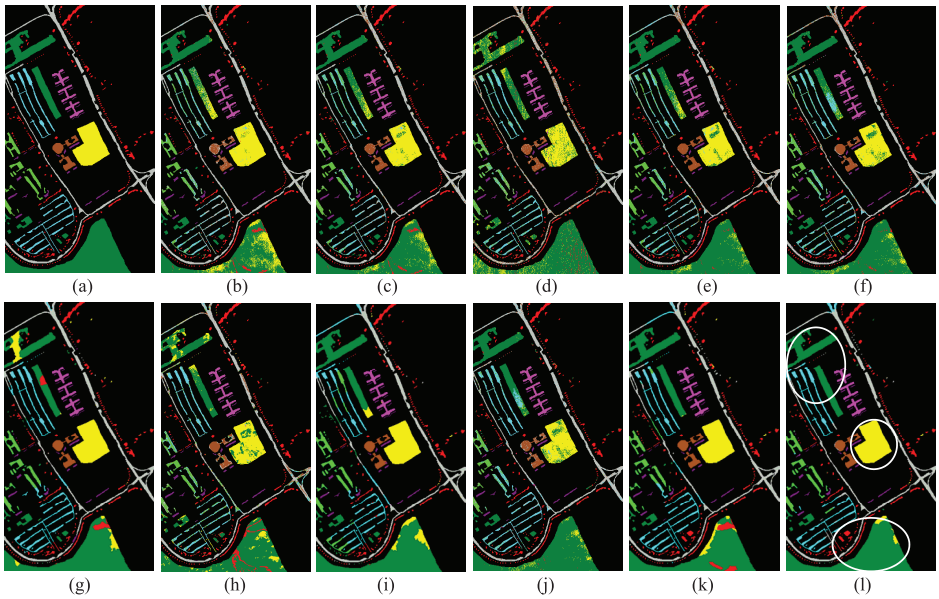


Fig. 10. Classification maps on the PaviaU dataset. (a) Ground-truth map. (b) DRCNN. (c) F3DCNN. (d) miniGCN. (e) NLGCN. (f) DRGCN. (g) MSSGU. (h) DGUnet. (i) SSAGU. (j) DHCN. (k) F2HNN. (l) CHGUnet.

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PAVIAU DATASET

Class	DRCNN	F3DCNN	miniGCN	NLGCN	DRGCN	MSSGU	DGUnet	SSAGU	DHCN	F2HNN	CHGUnet
1	79.44±1.35	79.39±0.91	70.63±2.55	87.11±1.01	92.37±0.45	90.71±0.73	88.12±0.94	93.75±0.38	81.78±1.57	91.78±0.49	**95.20±0.36**
2	89.31±0.63	90.30±0.57	82.23±1.16	89.49±1.02	92.04±0.69	92.94±0.67	88.98±0.89	88.42±1.01	93.66±0.39	90.55±0.45	**96.88±0.14**
3	82.16±0.89	86.59±0.96	86.32±0.62	76.33±1.29	73.74±1.79	93.86±0.34	65.06±2.73	91.82±0.66	82.62±1.39	91.27±0.84	**97.96±0.10**
4	93.29±0.34	94.20±0.46	90.24±0.55	95.72±0.34	90.68±0.39	85.31±1.26	87.89±1.17	88.76±0.45	94.74±0.08	89.83±0.80	**96.40±0.38**
5	99.84±0.01	99.36±0.04	99.32±0.03	97.99±0.12	**100±0.00**	99.32±0.03	99.61±0.02	99.46±0.03	**100±0.00**	98.56±0.10	**100±0.00**
6	82.68±0.79	88.66±1.01	80.86±1.09	89.62±0.83	87.85±1.01	**100±0.00**	86.16±1.14	**100±0.00**	90.72±0.83	99.92±0.01	**100±0.00**
7	93.81±0.47	91.22±0.73	91.77±0.55	87.11±1.05	85.03±1.02	99.16±0.04	84.61±0.83	**100±0.00**	88.05±0.66	**100±0.00**	97.98±0.13
8	75.55±1.67	83.72±1.60	71.47±2.04	83.27±0.74	84.42±0.77	88.82±1.95	76.73±1.10	97.76±0.09	97.69±0.18	**96.06±0.31**	92.52±0.99
9	**100±0.00**	99.65±0.03	**100±0.00**	99.78±0.01	99.89±0.01	95.06±0.40	**100±0.00**	94.93±0.34	**100±0.00**	99.67±0.02	**100±0.00**
OA	85.43±1.19	88.24±0.70	82.63±1.61	88.43±0.62	91.49±0.65	93.10±0.65	88.51±0.77	94.46±0.49	92.67±0.53	93.54±0.58	**97.28±0.36**
AA	87.45±0.98	89.21±0.52	86.20±0.98	89.34±0.85	90.55±0.49	93.82±0.54	87.82±0.82	95.05±0.33	91.13±0.82	95.29±0.34	**97.02±0.38**
Kappa	81.37±0.78	84.87±1.16	77.34±2.19	85.29±1.33	89.36±0.96	88.33±0.92	85.41±0.79	93.20±0.84	90.41±0.54	91.32±0.90	**95.41±0.55**

Notably, the proposed CHGUnet surpasses all comparison methods on most classes and achieves the highest OA, AA, and kappa coefficients on three datasets. Even compared with advanced models based on hypergraph convolution and graph U-Net (e.g. DHCN, F2HNN, MSSGU, and SSAGU), CHGUnet achieves OA improvements of 5.07%–1.65% on

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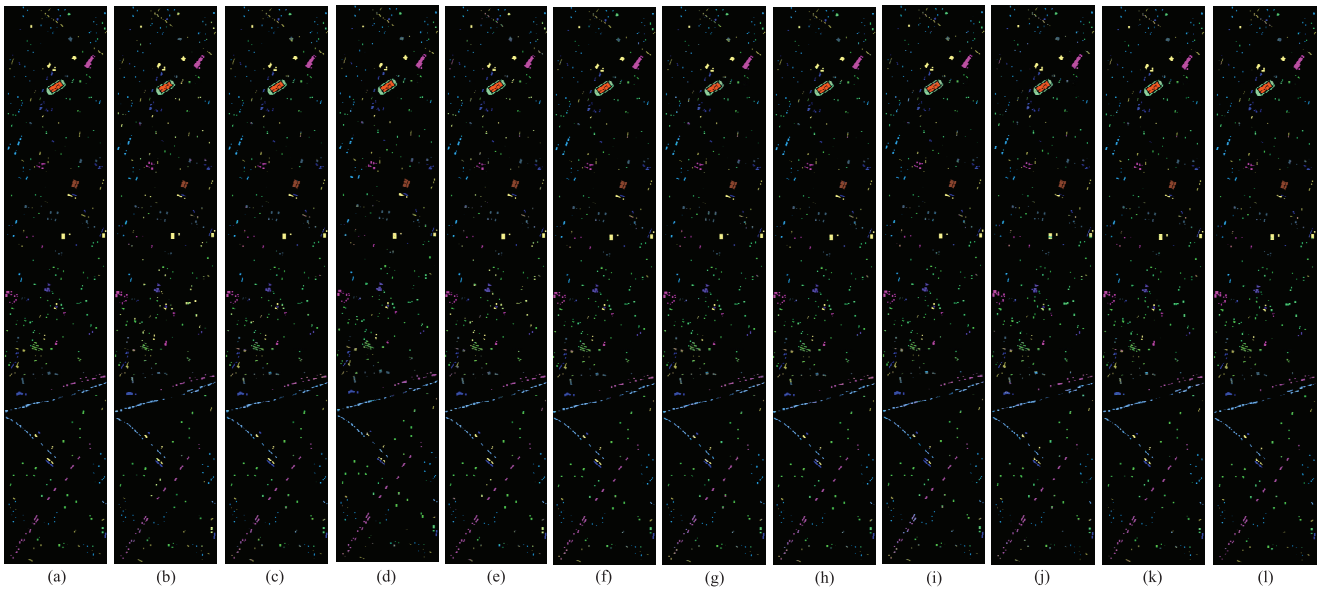


Fig. 11. Classification maps on the Houston dataset. (a) Ground-truth map. (b) DRCNN. (c) F3DCNN. (d) miniGCN. (e) NLGCN. (f) DRGCN. (g) MSSGU. (h) DGUnet. (i) SSAGU. (j) DHCN. (k) F2HNN. (l) CHGUnet.

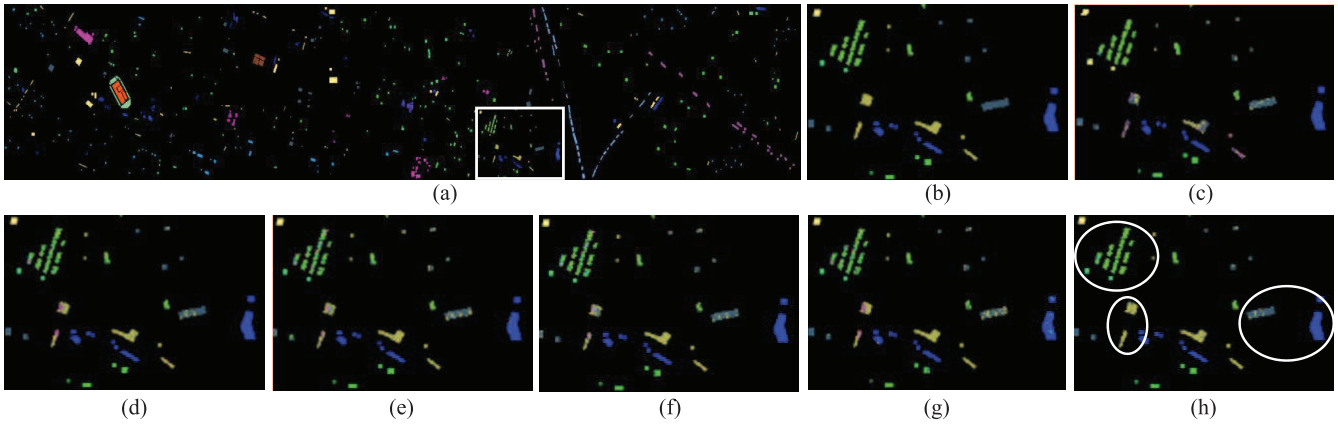


Fig. 12. Classification maps of typical area [white rectangular frame in (a)] on the Houston dataset with methods whose OA > 90%. (b) Ground-truth map. (c) NLGCN. (d) DRGCN. (e) MSSGU. (f) SSAGU. (g) F2HNN. (h) CHGUnet.

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET

Class	DRCNN	F3DCNN	miniGCN	NLGCN	DRGCN	MSSGU	DGUnet	SSAGU	DHCN	F2HNN	CHGUnet
1	99.64±0.03	99.95±0.02	98.67±0.19	99.59±0.05	97.50±0.31	94.54±0.84	95.05±0.47	99.69±0.05	91.67±0.64	**99.97±0.02**	98.78±0.19
2	98.34±0.13	**100±0.00**	99.86±0.01	99.53±0.04	99.65±0.05	98.07±0.22	97.92±0.25	98.99±0.12	96.26±0.37	97.53±0.17	99.31±0.05
3	83.02±2.31	97.33±0.36	90.76±1.36	99.58±0.04	96.63±0.47	100±0.00	98.16±0.25	99.38±0.10	98.45±0.13	99.68±0.05	**100±0.00**
4	99.70±0.02	99.48±0.05	99.62±0.05	**99.85±0.02**	99.63±0.05	97.13±0.25	96.33±0.43	99.18±0.11	95.64±0.56	98.11±0.18	**99.85±0.02**
5	98.14±0.22	99.11±0.12	98.29±0.18	97.82±0.31	97.3±0.25	96.79±0.39	98.77±0.15	99.01±0.12	95.97±0.50	98.88±0.11	**100±0.00**
6	99.62±0.05	99.41±0.09	99.74±0.02	99.66±0.05	97.67±0.31	97.74±0.29	97.78±0.25	**100±0.00**	96.36±0.26	98.37±0.16	**100±0.00**
7	95.41±0.70	93.65±0.86	95.43±0.08	99.61±0.06	92.26±0.58	91.39±0.60	86.04±1.10	**98.38±0.25**	87.93±1.93	92.17±0.89	96.66±0.32
8	72.54±3.16	75.15±3.41	71.38±4.62	83.90±1.59	61.71±4.95	86.23±1.76	79.11±2.42	84.59±2.55	74.54±3.55	85.05±1.97	**88.54±1.46**
9	91.55±0.93	**100±0.00**	99.01±0.10	98.91±0.13	99.17±0.11	88.34±1.20	78.65±3.24	99.15±0.06	77.79±3.12	86.18±0.96	96.84±0.36
10	91.20±1.18	85.99±0.96	91.02±1.22	95.37±0.35	90.12±1.38	88.77±0.82	80.04±3.08	91.14±1.36	92.78±0.53	95.39±0.72	**95.74±0.61**
11	98.41±0.08	95.27±0.47	98.43±0.17	97.18±0.42	89.88±1.58	87.57±1.44	80.70±1.81	97.25±0.35	90.55±1.44	90.74±1.13	**99.24±0.04**
12	96.38±0.28	98.27±0.19	98.72±0.19	94.41±0.05	81.84±2.99	84.48±1.33	75.99±2.10	**97.79±0.37**	85.46±2.33	83.61±2.49	95.33±0.77
13	**100±0.00**	99.19±0.08	97.37±0.38	99.76±0.03	97.23±0.40	86.50±1.07	59.08±5.04	93.08±0.83	73.22±4.42	88.38±1.18	94.04±0.52
14	97.55±0.21	92.88±0.61	93.63±0.53	95.45±0.53	93.92±0.76	96.47±0.31	99.39±0.08	97.73±0.26	99.47±0.08	**100±0.00**	**100±0.00**
15	80.83±2.43	79.27±3.09	60.20±6.08	70.83±2.51	83.66±2.61	98.50±0.12	98.21±0.19	73.45±3.90	98.36±0.24	99.31±0.05	**100±0.00**
OA	88.71±1.05	90.08±1.09	87.09±2.14	91.26±1.02	90.33±1.21	91.58±0.72	87.58±1.08	93.86±0.55	89.07±1.29	94.12±0.61	**96.26±0.56**
AA	93.84±0.44	94.72±0.82	93.40±0.78	95.09±0.40	91.88±0.56	92.83±0.51	88.08±1.92	95.22±0.56	90.29±0.82	95.79±0.69	**97.41±0.26**
Kappa	87.35±1.80	88.26±1.24	85.91±2.19	90.16±0.71	90.76±0.73	91.52±1.10	87.36±0.95	93.41±1.03	88.64±1.21	93.68±0.73	**95.82±0.35**

Indian Pines dataset, 4.61%–2.82% on PaviaU dataset, and 7.19%–2.14% on the Houston dataset. These gains primarily stem from two key innovations: 1) The novel high-order topology-aware pooling and unpooling operations enable

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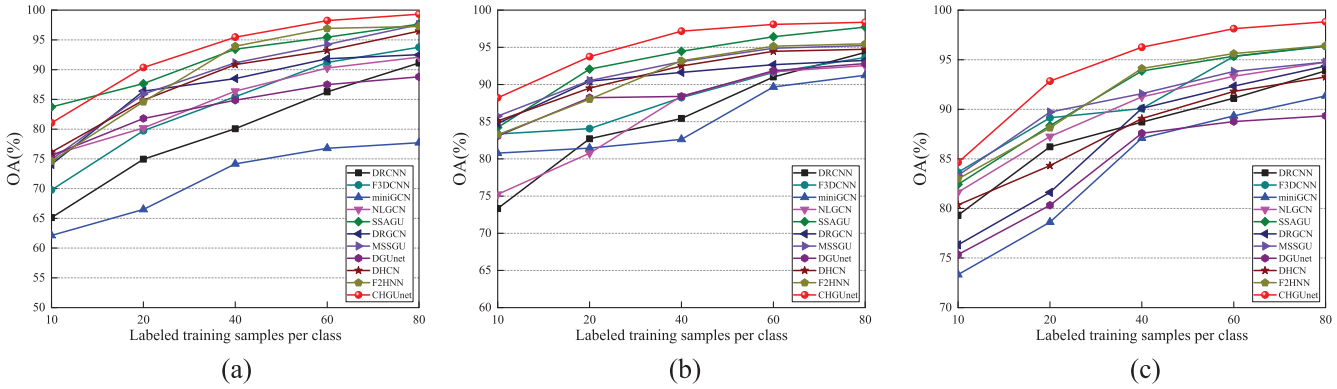


Fig. 13. OAs with different numbers of the labeled training samples. (a) Indian Pines. (b) PaviaU. (c) Houston.

more effective feature aggregation in deep HGCN networks while mitigating over-smoothing, addressing a fundamental limitation in existing hypergraph methods and 2) the integration of pixel-level depthwise separable convolution with superpixel-level hypergraph convolution through cross-level fusion captures both local details and global structural relationships that pure graph U-Net-based approaches often miss. This classification performance is also qualitatively confirmed by visual classification maps in Figs. 9–12. For classification maps on Indian Pines and PaviaU datasets, it can be clearly seen that the maps of CHGUnet have the weakest misclassification situation and are closest to the ground-truth map, especially in areas marked with white circles. Because the classification targets are small and scattered in the Houston dataset, we enlarge the same typical area in the classification maps of the models whose OA is higher than 90% for comparison in Fig. 12. In these enlarged maps, CHGUnet still shows a more accurate classification effect than other advanced GUnet-based or HGCN-based models even in the face of scattered small ground objects.

E. Analysis of Training Samples

In previous experiments, we have shown the results of CHGUnet and other methods under fixed training sample conditions. Thereupon, we set the number of labeled training samples per class to 10, 20, 40, 60, and 80 to explore the performance of CHGUnet under different training conditions. As the results shown in Fig. 13, all methods have improved in OAs with the increase in labeled training samples because the larger labeled training set containing richer prior information will lead to an implicit increase in the discriminative power of deep networks. Overall, the proposed CHGUnet outperforms other CNN-based and HGCN-based methods, and this phenomenon is particularly evident in the Houston and PaviaU datasets. Compared with the DHCN and F2HNN method based on hypergraph, CHGUnet still gets the best accuracies across all datasets even under the smallest labeled training condition. Its strong learning capability on small training samples benefits from the powerful topology representation ability of clustering hypergraph and structural optimization capability of HGPool.

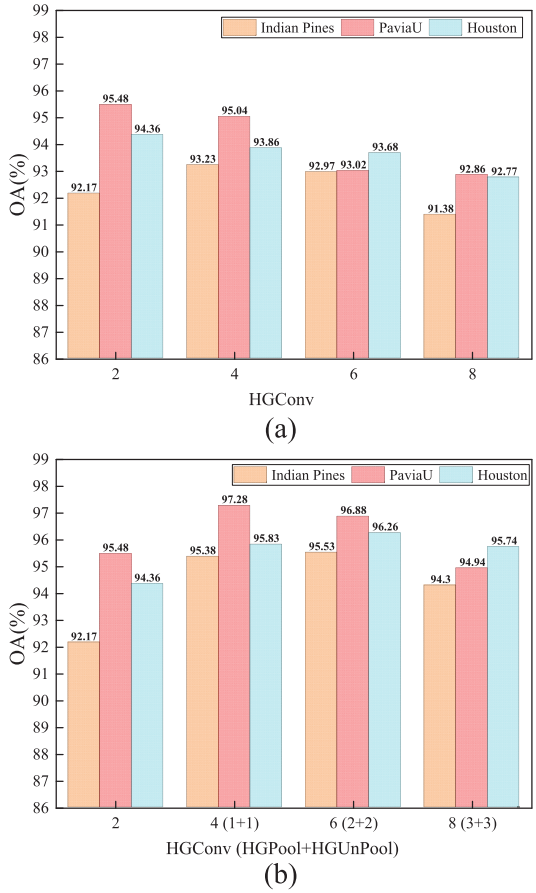


Fig. 14. OAs with different network depth. (a) Without HGPool and HGUnPool layers. (b) With HGPool and HGUnPool layers.

F. Investigation of HGPool and HGUnPool Operations

HGPool and HGUnPool operations are the core of establishing the hypergraph U-Net architecture. In order to explore their ability to eliminate the over-smoothing phenomenon and enhance the effective learning of structured features in deep networks, we set the number of HGConv layers in CHGUnet in the range of $\{2, 4, 6, 8\}$, matching or not matching the HGPool and HGUnPool layers, respectively. The corresponding results on three datasets are exhibited in Fig. 14.

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT FEATURE FUSION METHODS [OA ± STD (%)]

Data sets	ADD	AFF	CAFM	CLIF
Indian Pines	91.45±1.03	92.86±1.16	94.03±0.74	95.53±0.59
PaviaU	93.86±0.88	94.19±0.80	96.45±0.94	97.28±0.36
Houston	93.94±0.79	94.56±1.34	94.67±0.86	96.26±0.56

In Fig. 14(a), when there is no adaptation HGPool and HGUnPool layers, the classification accuracies reach peak values when the number of HGConv layers is 4 in the Indian Pines, and when the number of layers is 2 in the PaviaU and Houston datasets. Then, the accuracies decrease to varying degrees as the number of HGConv layers increases. When the hypergraph U-Net architecture lacks pooling and unpooling layers, it becomes a multilayer hypergraph convolutional network based on a fixed topology structure. Thus, it cannot dynamically adjust the hypergraph structure according to the node states in different layers to eliminate redundant high-order topological connections, resulting in over-smoothing node features and suppressing the learning ability of deep networks. Fig. 14(b) further discusses the network with different numbers of HGPool and HGUnPool layers. The introduction of HGPool and HGUnPool significantly enhances the classification accuracy, especially in deeper network architecture. By comparing Fig. 14(a) and (b), we can find that the performance of the networks with the same depth has generally been improved by integrating HGPool and HGUnPool operations. These improvements are thanks to the layer-by-layer optimization of the hypergraph with the aid of the high-order topology-aware pooling from the dual views of node and hyperedge.

G. Investigation of Fusion Strategy

CHGUnet captures structural features at the superpixel level while the lightweight depthwise separable branch extracts fine-grained semantic features at the pixel level. The CLIF is a crucial step in obtaining the final complementary feature representations of HSI. To confirm the effectiveness of CLIF, we compare it with other famous attention-based fusion methods, including attention feature fusion (AFF) [36], cross attention fusion mechanism (CAFM) [37] and basic addition (ADD) fusion methods. The classification results of CHGUnet combining with different fusion methods, are given in Table VII.

AFF and CAFM fusion methods have absolute advantages over coarse addition fusion, because both methods use an attention mechanism to adjust the pixel-level and superpixel-level features and enable the fused features to retain their respective valuable information. Compared with AFF and CAFM, CLIF exhibits more advanced performance owing to its two distinct advantages: 1) comprehensive channel relationship modeling: CLIF considers all interchannel correlations across different feature levels, enabling more accurate weight assignment and 2) balanced feature contribution: the pair of complementary attention weights obtained from CLIF prevents any feature being completely submerged by the huge scale differences between superpixel-level and pixel-level features.

H. Ablation Analysis

For the proposed CHGUnet, besides CLIF, HGPool and HGUnPool, there are some critical components, including SFC for hypergraph construction, HGConv operation, and DFGA used in HGUnPool. The CLIF module and a pair of HGPool and HGUnPool operations have already been discussed in the previous subsections. Thus, we analyze the contributions of the other three crucial components (SFC, HGConv, and DFGA) on three datasets. The ablation experimental results are presented in Table VIII, where symbol ✓ indicates the implementation of the corresponding component. The ablation versions V1–V3 explore the impact of SFC and DFGA by removing them from CHGUnet. When verifying SFC, we replace it with a common k-nearest neighbor (KNN) selection method for hypergraph construction. For validating HGConv using high-order topology, we replace the hypergraph in CHGUnet with pairwise graph built on KNN as version V4. In addition, relying on the clique expansion (CE) technology [38], we convert high-order hypergraph into simple graph by connecting all nodes in each hyperedge pairwise, and then build the GConv-based variant versions V5 and V6 of CHGUnet, which provide an alternative way to realize hypergraph convolution and pooling operations via the mature related operations on simple graphs.

According to the results of different ablation versions V1–V6 reported in Table VIII, the performance gets worse when some components are removed or replaced from the complete CHGUnet. Version V1 shows the poorest classification performance, as it removes both SFC and DFGA for degenerating into a simple stacked convolutional network operating on a KNN-based hypergraph. Ablation studies of V2 and V3 demonstrate that the losing either of SFC or DFGA will result in pronounced performance degradation, which indicates each proposed component contributes to the performance improvement of CHGUnet. For versions V4–V6, they mainly replace HGConv with regular GConv to explore the impact of hypergraph on deep structure feature learning. The lower classification accuracy of V4 compared with V5 demonstrates that hypergraph-to-graph conversion via CE preserves more comprehensive topological information than direct KNN graph construction. The experimental results demonstrate significant advantages of the original CHGUnet that conducts the high-order pooling on hypergraph directly, outperforming the V6 based on hypergraph CE by 1.61%–5.36% in OAs across all datasets. While CE preserves node connections of the original hypergraph, it leads to substantial edge explosion and fails to maintain the natural high-order structure among multiple nodes. Meanwhile, the standard graph pooling operations applied to the hypergraph CE-based graph cannot effectively handle these dense pairwise connections and capture the multinode relationships. Furthermore, comparisons between V5 and V6 or V3 and CHGUnet reveal that removing DFGA consistently reduces classification performance no matter whether in hypergraph or graph U-Net architecture. This demonstrates the positive impact of adjusting hyperedge or edge weights through deviation feature guidance attention between paired pooling and unpooling layers.

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TABLE VIII
ABLATION RESULTS OF EACH COMPONENT ON THREE DATASETS. [OA ± STD (%) (KAPPA)]

Components	V1	V2	V3	V4	V5	V6	CHGUnet
SFC	—	—	✓	—	✓	✓	✓
KNN	✓	✓	—	✓	—	—	—
CE	—	—	—	—	✓	✓	—
HGConv	✓	✓	✓	—	—	—	✓
GConv	—	—	—	✓	✓	✓	—
DFGA	—	✓	—	—	—	✓	✓
Indian Pines	88.20±1.66 (0.85)	92.86±1.02 (0.91)	93.39±0.95 (0.92)	90.86±0.74 (0.87)	91.49±0.84 (0.90)	91.56±1.22 (0.90)	95.53±0.59 (0.94)
PaviaU	87.87±1.24 (0.84)	94.47±0.76 (0.93)	93.82±0.70 (0.92)	88.54±0.91 (0.85)	90.33±1.06 (0.88)	91.92±0.97 (0.90)	97.28±0.36 (0.95)
Houston	87.28±1.06 (0.87)	93.85±0.60 (0.93)	93.36±0.69 (0.93)	89.68±0.75 (0.89)	91.29±0.59 (0.91)	94.65±0.92 (0.94)	96.26±0.56 (0.96)

IV. CONCLUSION

This article presents a novel CHGUnet for HSI classification, which is mainly composed of hypergraph U-Net architecture with a pair of designed HGPool and HGUnPool operations. Based on the awareness of multivariate relationships among related samples in SFC, a clustering hypergraph representing the high-order structure among multiple superpixel samples is built as the backbone of the network. Then, a hierarchical U-Net architecture is developed for hypergraph to learn the deep high-order structural features of HSI data. During the encoding stage of U-Net, the topological information of both nodes and hyperedges is simultaneously considered to realize the HGPool operation, which allows the hypergraph to remain a more valuable topological association after pooling. Meanwhile, to restore hypergraph structure in the decoding stage, the HGUnPool layer has been developed on DFGA for recovering hypergraph details and optimizing hyperedge weights to adapt to the current node state. Finally, a lightweight auxiliary branch of pixel-level feature extraction is integrated into superpixel hypergraph U-Net through a CLIF mechanism, thus implementing better coarse and fine-grained information matching to enhance the feature complementarity. The experimental results on three HSI datasets have proven the superior performance of CHGUnet.

REFERENCES

[1] J. Qu, X. Wu, W. Dong, J. Cui, and Y. Li, "IR&ArF: Toward deep interpretable arbitrary resolution fusion of unregistered hyperspectral and multispectral images," *IEEE Trans. Image Process.*, vol. 34, pp. 1934–1949, 2025.

[2] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[3] X. Kang, Y. Zhu, P. Duan, and S. Li, "Two-dimensional spectral representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502809.

[4] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.

[5] L. Zhang, L. Song, B. Du, and Y. Zhang, "Nonlocal low-rank tensor completion for visual data," *IEEE Trans. Cybern.*, vol. 51, no. 2, pp. 673–685, Feb. 2021.

[6] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[7] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

[8] Z. Gong, P. Zhong, and W. Hu, "Statistical loss and analysis for deep learning in hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 322–333, Jan. 2021.

[9] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[10] A. Sellami, A. Ben Abbes, V. Barra, and I. R. Farah, "Fused 3-D spectral–spatial deep neural networks and spectral clustering for hyperspectral image classification," *Pattern Recognit. Lett.*, vol. 138, pp. 594–600, Oct. 2020.

[11] J. Qu, W. Dong, Q. Du, Y. Yang, Y. Xu, and Y. Li, "Cyclic consistency constrained multiview graph matching network for unsupervised heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605315.

[12] K. Wu, G. Xu, Y. Zhang, and B. Du, "Hyperspectral image target detection via integrated background suppression with adaptive weight selection," *Neurocomputing*, vol. 315, pp. 59–67, Nov. 2018.

[13] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[14] F. F. Shahraki and S. Prasad, "Graph convolutional neural networks for hyperspectral data classification," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 968–972.

[15] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[16] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[17] R. Chen, G. Li, and C. Dai, "DRGCN: Dual residual graph convolutional network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] Y. Lu, S. Mei, F. Xu, M. Ma, and X. Wang, "Separable deep graph convolutional network integrated with CNN and prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516216.

[19] Y. Yang, D. Li, H. Wang, Y. Feng, L. Yan, and G. Zhang, "Hyperspectral image classification using a residual enhanced feature fusion hypergraph neural network," *Remote Sens. Lett.*, vol. 15, no. 3, pp. 313–325, Mar. 2024.

[20] Y. Chu et al., "Hyperspectral image classification using feature fusion fuzzy graph broad network," *Inf. Sci.*, vol. 689, Jan. 2025, Art. no. 121504.

[21] H. Cao, J. Cao, Y. Chu, Y. Wang, G. Liu, and P. Li, "Global-local manifold embedding broad graph convolutional network for hyperspectral image classification," *Neurocomputing*, vol. 602, Oct. 2024, Art. no. 128271.

[22] J. Xu, J. E. Fowler, and L. Xiao, "Hypergraph-regularized low-rank subspace clustering using superpixels for unsupervised spatial–spectral hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 871–875, May 2021.

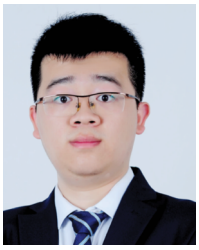
[23] Z. Ma and B. Yang, "Spatial–spectral hypergraph-based unsupervised band selection for hyperspectral remote sensing images," *IEEE Sensors J.*, vol. 24, no. 17, pp. 27870–27882, Sep. 2024.

[24] Z. Ma, Z. Jiang, and H. Zhang, "Hyperspectral image classification using feature fusion hypergraph convolution neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517314.

[25] Q. Xu, S. Xu, J. Liu, and L. Huang, "Dynamic hypergraph convolution and recursive gated convolution fusion network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

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[26] H. Gao, Y. Liu, and S. Ji, "Topology-aware graph pooling networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 12, pp. 4512–4518, Dec. 2021.
[27] D. Mesquita, A. H. Souza, and S. Kaski, "Rethinking pooling in graph neural networks," in *Proc. IEEE Int. Conf. Neural Inf. Process. Syst.*, Jun. 2020, pp. 2220–2231.
[28] F. Bianchi, D. Grattarola, and C. Alippi, "Spectral clustering with graph neural networks for graph pooling," in *Proc. Int. Conf. Mach. Learn.*, Nov. 2020, pp. 874–883.
[29] Z. Zhang et al., "Hierarchical multi-view graph pooling with structure learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 545–559, Jan. 2023.
[30] X. Zhang, S. Chen, P. Zhu, X. Tang, J. Feng, and L. Jiao, "Spatial pooling graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521315.
[31] F. Guo, Z. Li, Z. Xin, X. Zhu, L. Wang, and J. Zhang, "Dual graph U-Nets for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8160–8170, 2021.
[32] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "Multilevel superpixel structured graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516115.
[33] K. Qu, C. Wang, Z. Li, and F. Luo, "Spatial–spectral attention graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528317.
[34] K.-S. Choi and K.-W. Oh, "Fast simple linear iterative clustering for superpixel segmentation," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2015, pp. 552–553.
[35] J.-G. Jang, C. Quan, H. D. Lee, and U. Kang, "FALCON: Lightweight and accurate convolution based on depthwise separable convolution," *Knowl. Inf. Syst.*, vol. 65, no. 5, pp. 2225–2249, May 2023.
[36] Y. Dai, F. Gieseke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3560–3569.
[37] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.
[38] Y. Zhou, A. Rathore, E. Purvine, and B. Wang, "Topological simplifications of hypergraphs," *IEEE Trans. Vis. Comput. Graph.*, vol. 29, no. 7, pp. 3209–3225, Jul. 2023.



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